3.7 Solving Nonlinear Systems of Equations

Name _____ Period ____ Date _____

Algebra II 3.7 (30-minute plan) · Student sheet

I can…

- solve a system with a linear and a quadratic equation by graphing.
- solve nonlinear systems by substitution and elimination.
- solve f(x) = g(x) by graphing or algebra.

Vocabulary · fill in the missing word

System of nonlinear equations — _____

Line and parabola — can _____ in 0, 1, or 2 points

Two parabolas — can meet in 0, 1, 2, or _____ many points

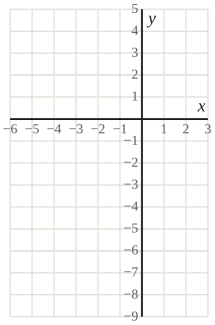
f(x) = g(x) — the _____ where the graphs of y = f(x) and y = g(x) meet

Part 1 · Notes and examples (fill in each note, then work the example)

Line and parabola: A line and a parabola can meet in _____ points.

Example 1

Solve the system by graphing. $y = x^2 + 4x - 4$ and $y = 2x - 5$



Substitution: Substitution: solve one equation for _____, then _____.

Both coordinates: Always find _____ coordinates of each solution.

Example 2

Solve the system. $y = x^2 - 1$ and $y = x + 1$

Example 3

Solve the system.

$x^2 - 4x - y = -5$

$x^2 - 2x + y = 5$

Part 2 · Partner practice

Partner practice

Solve the system. $y = x^2$ and $y = -x + 6$